AIH5571 Introduction to Generative AI and Its Applications

Assessment Rubrics for Major Assessment Tasks

Subject Information

Subject Code	AIH5571
Subject Title	Introduction to Generative AI and Its Applications
Credit Value / Level	3 Credits / Level 5
Subject Objectives	The subject is envisioned for students to embrace and leverage the potential of Generative Artificial Intelligence (AI). It will primarily focus on the nature, development, and implications of Generative AI, its role in innovative research methods, its application across various disciplines, and its ethical and societal implications.
Intended Learning Outcomes (ILOs)	Upon completion of the subject, students will be able to: • **(a)** Demonstrate an understanding of the foundational concepts of Generative AI; • **(b)** Acquire practical skills in using Generative AI technologies; • **(c)** Demonstrate an awareness of global contemporary ethics issues and the impact of Generative AI applications in daily life.
Assessment Breakdown	• Exercises and Assignments: 15% (ILO b) • Quizzes: 10% (ILOs a, b, c) • Essay: 15% (ILOs a, c) • **Course Project: 60% (ILOs a, b, c) — Major Assessment Component**

Course Project Assessment Structure

Assessment Deliverable	Project Weighting	Overall Subject Weighting	Target Intended Learning Outcomes
1. Project Proposal	10%	6%	ILO (a), ILO (b)
2. Final Project & Technical Report	70%	42%	ILO (a), ILO (b), ILO (c)
3. Oral Presentation & Demonstration	20%	12%	ILO (a), ILO (b), ILO (c)
Total Course Project	**100%**	**60%**	**All ILOs**

1. Rubrics for Project Proposal (10% of Project / 6% of Subject)

Students submit a project proposal outlining the problem statement, background, proposed Generative AI methodology/tools, and work plan.

Component & Weight	Excellent (A+, A, A-)	Good (B+, B, B-)	Satisfactory (C+, C, C-)	Pass (D+, D)	Failure (F)
Project Objective Formulation, Background & GenAI Methodology (60%) *[ILO a, b]*	The objectives and use cases are well defined, rigorous, and prioritized. Relevant GenAI technologies, foundation models, and system constraints are thoroughly obtained and accurately analyzed. Technical approach, prompt	All major project objectives are identified. Sufficient background information on the target problem and GenAI domain is obtained. Appropriate model choices and methodologies are selected. Design recommendations are reasonable and	Most major objectives are identified, but some specific scopes are vague or priorities are not established. Relevant GenAI constraints are partially addressed. The proposed methodology lacks technical depth or relies on generic approaches	Objectives are poorly formulated or barely defined. Background is superficial. Proposed solution ignores key GenAI constraints or technical alternatives.	Major objectives are unfeasible, missing, or fundamentally flawed. Demonstrates no understanding of GenAI concepts or required tools.

Component & Weight	Excellent (A+, A, A-)	Good (B+, B, B-)	Satisfactory (C+, C, C-)	Pass (D+, D)	Failure (F)
	engineering framework, and design recommendations are exceptionally well supported by literature and practical feasibility.	mostly supported by sound reasoning.	without tailored justification.		
Clarity and Presentation of Proposal Report (30%)	The proposal is exceptionally well organized, professional, and clearly written. The architectural and conceptual logic is articulated seamlessly. Language is fluent, precise, and free from grammatical errors. Diagrams, pipeline charts, and tables effectively enhance comprehension.	The proposal is well organized and clearly written for the most part. The technical flow is easy to follow with minor ambiguities. Professional terminology is used appropriately. Diagrams are consistent with the text. Language is grammatical with few minor errors.	The report is organized by basic sections, but the flow of ideas is disjointed or difficult to follow in places. Diagrams or workflows are incomplete or poorly explained. Several grammatical or structural errors impede smooth communication.	Structure is weak and disjointed. Significant effort required to follow the technical logic. Diagrams are unclear or missing. Frequent grammatical errors.	Lacks coherent structure and organization. Writing is fragmented, unreadable, or severely incomplete.
Planning of Future Work & Labor Allocation (10%)	Comprehensive, realistic, and detailed project milestone schedule with well-defined deliverables. Clear risk assessment and mitigation	Well-defined milestone schedule and good plan of future implementation work. Logical timeline with practical labor allocation	Basic work plan and task list provided, but timeline lacks detail or realistic milestones. Work allocation is vaguely defined or	Timeline is unrealistic or lacks specific milestones. Task allocation among members is minimally defined or heavily	No clear work plan or milestone schedule provided. Labor allocation is completely absent.

Component & Weight	Excellent (A+, A, A-)	Good (B+, B, B-)	Satisfactory (C+, C, C-)	Pass (D+, D)	Failure (F)
	strategies. Task allocation among team members is balanced, practical, and clearly aligned with individual strengths.	across team members. Minor omissions in contingency planning.	unbalanced across team members.	unbalanced.	

2. Rubrics for Final Project & Technical Report (70% of Project / 42% of Subject)

Students submit a working Generative AI application / pipeline along with comprehensive source code, prompts, configurations, and a comprehensive final technical report.

Component & Weight	Excellent (A+, A, A-)	Good (B+, B, B-)	Satisfactory (C+, C, C-)	Pass (D+, D)	Failure (F)
Results Obtained & Implementation Quality (60%) *[ILO b]*	Delivers an exceptionally high-quality, fully functional GenAI application or workflow. Demonstrates mastery of prompt engineering, model integration, or fine-tuning/RAG pipelines. Executes a comprehensive suite of qualitative and quantitative test cases with rigorous	Delivers a fully working GenAI application meeting all specified core functional requirements. Systematic prompt design and solid model integration. Executes a clear list of test cases with expected results verified. Code and pipeline configurations are structured and clean.	Delivers a partially functioning application where core features work but some modules fail or malfunction under specific inputs. Prompt strategies are basic and unrefined. Test cases are limited, covering only simple/ideal scenarios.	Delivers a minimally working or severely limited system. Only simplest baseline features execute. Code is poorly structured or untested.	Fails to deliver a functioning GenAI application. System is broken, non-executable, missing, or does not follow specifications.

Component & Weight	Excellent (A+, A, A-)	Good (B+, B, B-)	Satisfactory (C+, C, C-)	Pass (D+, D)	Failure (F)
	error handling and validation. Exceeds standard design specifications with innovative enhancements.				
Clarity, Technical Documentation & Ethical Impact Analysis (30%) *[ILO a, c]*	Report is thoroughly structured and articulated with exceptional scholarly rigor. Comprehensive evaluation of foundational concepts. In-depth, nuanced critical analysis of ethical, legal, and societal implications (e.g., bias, privacy, hallucination mitigation, safety, intellectual property). Flawless academic writing and documentation.	Report is well organized and clearly written. Demonstrates good understanding of GenAI concepts. Includes a well-grounded analysis of ethical and societal implications in the context of the project. Clear technical documentation with minimal errors.	Report is moderately organized but contains gaps in technical explanation. Discussion of ethical and societal implications is superficial, generic, or fragmented. Noticeable grammatical or formatting shortcomings.	Report is poorly written with major gaps in technical explanation. Ethical analysis is minimal, purely descriptive, or lacks critical reflection.	Report is disjointed, severely incomplete, or incoherent. Ethical and societal impact analysis is completely absent or irrelevant.
Use of GenAI Engineering & System Design	Employs advanced software and	Employs appropriate software	Employs basic design techniques with	Demonstrates rudimentary or unorganized	Incoherent architecture and inability to

Component & Weight	Excellent (A+, A, A-)	Good (B+, B, B-)	Satisfactory (C+, C, C-)	Pass (D+, D)	Failure (F)
Techniques (10%) *[ILO a, b]*	AI engineering techniques (e.g., modular architecture, robust API/model orchestration, structured output parsing, versioning, context caching). Proposes creative and optimal technical solutions to project challenges.	engineering and GenAI system design practices. Demonstrates solid competency in API usage, prompt templates, and data pipeline management to successfully complete the project.	limited modularity. Progress is made toward technical challenges, but pipeline design is suboptimal or fragile.	system design. Pipeline components are loosely coupled with weak technical execution.	manage technical components. Fails to apply basic engineering practices.

3. Rubrics for Oral Presentation & Demonstration (20% of Project / 12% of Subject)

Each group presents their project outcomes followed by a live demonstration of their working GenAI solution and a Q&A session.

Component & Weight	Excellent (A+, A, A-)	Good (B+, B, B-)	Satisfactory (C+, C, C-)	Pass (D+, D)	Failure (F)
Project & System Demonstration (50%) *[ILO b]*	Demonstrates a fully functioning, robust GenAI application with inventive and original features. Smoothly showcases live interactions,	Demonstrates a working product supporting all core desired functions. Effectively shows main workflows and model responses during the live	Demonstrates a working product, but some features malfunction or fail during the live session. The demo is basic, simply walking through default inputs without	Application barely runs or experiences frequent crashes/errors during the demo. Shows limited preparation to showcase a working prototype.	System fails to run or crashes entirely. Unable to demonstrate any functional deliverable or working prototype.

Component & Weight	Excellent (A+, A, A-)	Good (B+, B, B-)	Satisfactory (C+, C, C-)	Pass (D+, D)	Failure (F)
	handling edge cases gracefully. Demonstration technique is imaginative, highly engaging, and clearly highlights key technical breakthroughs.	demo. Provides clear insights into the practical utility of the tool.	exploring system capabilities.		
Delivery, Visual Slides & Q&A (50%) *[ILO a, c]*	Slides are professionally designed with high-impact visual diagrams, benchmark charts, and concise text. Presentation delivery is fluent, confident, and engaging with excellent pacing within the time limit. Answers all questions raised in the Q&A session with deep technical insight and authoritative reasoning.	Slides provide an accurate and clear overview of key outcomes. Delivery is fluent and audible with good pacing and time management. Answers most questions in the Q&A session accurately and competently.	Slides are text-heavy or lack clear visual aids. Delivery is hesitant with noticeable pacing issues (over/under time). Answers to Q&A questions are partial, vague, or lack confidence.	Slides are disorganized or poorly formatted. Delivery lacks confidence and eye contact. Answers to basic Q&A questions are weak or confused.	Slides are unreadable or missing. Presentation is severely disorganized. Unable to answer basic questions during the Q&A session.

Institutional Policies & Submission Guidelines

1. Academic Honesty and Plagiarism Policy

The Hong Kong Polytechnic University takes a very serious view against academic dishonesty and plagiarism in all forms of student work, including continuous assessments, project code, and written reports. Plagiarised or fraudulent work will lead to disciplinary actions, including mark deduction, subject failure, or disciplinary proceedings as set out in the Student Handbook.

2. Policy on the Use of Generative AI Tools

Students are encouraged to embrace and responsibly leverage Generative AI tools in their project development. However, transparency and integrity are paramount:

- A formal **GenAI Declaration Form** must be completed and submitted alongside every assessment deliverable (Proposal, Final Report, Source Code).
- Students must explicitly document and cite the specific tools used, prompt strategies applied, and the nature/extent of AI contributions versus original student implementation.
- The University policy on GenAI usage is available at: https://www.polyu.edu.hk/ar/students-in-taught-programmes/use-of-genai/

3. Late Submission Policy

To ensure fairness, late submissions will incur a penalty of **10% deduction per 24-hour period** past the deadline. Submissions will **not be accepted after 3 calendar days (72 hours)** following the deadline, resulting in a zero mark for the late component.